MD. Mehedi Hasan Maruf

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Summary

Software engineer with 3+ years of experience building backends, web apps, and AI features in TypeScript and Python. I lead a small team and build TypeScript microservices and a retrieval-augmented generation (RAG) feature. I wrote Django and Node backends and looked after the production Linux servers they ran on. On the side I am building Omnious, a visual debugger that maps runtime errors onto a program's code graph.

Experience

Software Engineer II

Jan 2025 – Present

Aamira Software Solutions, Kokomo, Indiana, USA (remote)

- Lead three junior engineers on the company's React applications, after starting on its Angular project.
- Raised client response rates by 30% by leading development of a CRM system.
- Cut API response time by 40% by designing and building a TypeScript microservice backend.
- Made the application 60% faster by optimizing its Supabase (PostgreSQL) queries.
- Building a retrieval-augmented generation (RAG) feature for the platform.

Junior Software Engineer

Nov 2022 – Apr 2024

Code Studio, Rajshahi, Bangladesh

- Kept 12 Linux servers at 99.8% uptime, and cut deployment time by 40% by reworking the CI/CD pipelines.
- Moved 5 client environments onto Docker, which made them simpler to deploy.
- Wrote JavaScript, TypeScript, and Python (Django) backends used by thousands of people a day.

Industrial Trainee

Mar 2025

Brain Station 23, Dhaka, Bangladesh

- Represented a 15-person trainee cohort to management.
- Built and deployed StudentAI (see Projects), an AI learning assistant on FastAPI and Next.js.

Selected Projects

Omnious | Visual Debugging Platform

Nov 2025 – Present

- Building a debugger that parses a TypeScript or JavaScript codebase into a graph of files, functions, and classes, then animates runtime traces over it and shows where errors cluster.
- An AI assistant suggests root causes tied to specific graph nodes, using embeddings in pgvector and a locally run LLM (Ollama). Community detection groups the graph into architectural layers.
- A TypeScript Turborepo monorepo: Fastify 5 and tRPC v11 backend, Next.js 16 / React 19 frontend, and PostgreSQL 15 on Supabase (pgvector, ltree, pg_trgm, row-level security).
- Ships a CLI indexer and Express and Next.js SDKs; runs in Docker Compose and deploys through GitHub Actions.

StudentAI | Intelligent Learning Companion

Mar 2025

- An AI learning assistant (FastAPI, Next.js) built during industrial training at Brain Station 23.
- Answers questions about uploaded documents through a LangChain RAG pipeline with semantic retrieval and conversation memory; it also analyzes YouTube lectures and writes quizzes.
- Accepts PDFs, images, and video, and has real-time WebSocket chat, JWT login, and progress tracking.
- Deployed on Vercel and Render with a Supabase database.

Skills

AI/ML: LLM applications, RAG, LangChain, Hugging Face Transformers, PyTorch, NLP, vector search (pgvector), model evaluation (RLHF)
Languages: TypeScript, Python, JavaScript, C++
Backend: NestJS, FastAPI, Fastify, tRPC, Django, .NET
Frontend: Next.js (React), Angular
Databases: PostgreSQL (Supabase), MongoDB, MySQL
DevOps: Docker, GitHub Actions (CI/CD), Linux server administration

Education

B.Sc. Engineering, Computer Science and Engineering *Mar 2022 – Aug 2026*

Rajshahi University of Engineering & Technology (RUET), Rajshahi, Bangladesh

- CGPA: 3.75 / 4.00, Honours
- Thesis: *TRACE*, which ranks the courses that hold students back using prerequisite graphs, Bayesian fail-rate models, and student-feedback sentiment, then checks the result with causal inference on 23,369 student records. A follow-up paper is under review at LAK 2027.

Certifications & Achievements

- Digital Accessibility Training (DAT) — a2i (Aspire to Innovate), Govt. of Bangladesh (Oct 2025)
- Industrial Training Certificate — Brain Station 23 (Mar 2025)
- PHP (Laravel) — EDGE Project (Feb 2025)
- Experience Certificate — Code Studio (Apr 2024)
- 4th Place — RUET CodeSmash, team competitive programming contest (Mar 2023)
- 4th Place — RUET CodeSpark, individual competitive programming contest (Oct 2022)